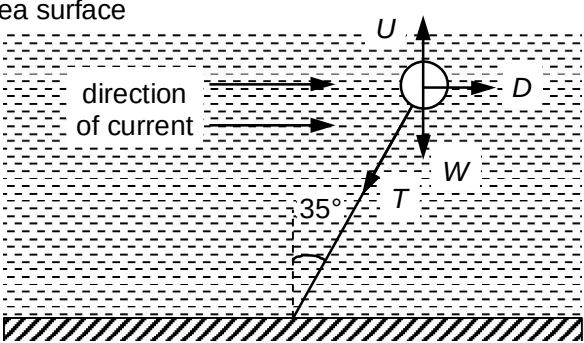


No.	Solution	Remarks
1(a)(i)	Upthrust is the <u>net upward force</u> acting on a body submerged in a fluid due to the <u>difference in pressure at the top and bottom</u> of the immersed portion of the body.	[1] for correct answer
1(a)(ii)	Upthrust $U = \frac{4}{5} V \rho_{\text{sea}} g$ $= \frac{4}{5} (7.5 \times 10^{-2})(1.03 \times 10^3)(9.81)$ $= 606.3$ $\approx 606 \text{ N}$	[1] for correct equation and numerical substitution [1] for correct answer
1(a)(iii)	$U = T + mg$ $T = U - mg$ $= 606.3 - (8.0)(9.81)$ $= 527.8$ $= 530 \text{ N (shown)}$	[1] for correct method and numerical substitution
1(b)(i)	 sea surface direction of current 35° sea bed [1] for correct T, U and D	
1(b)(ii)	Resolving forces horizontally, $T \sin 35^\circ = D \quad \text{--- (1)}$ Resolving forces vertically, $T \cos 35^\circ + W = U$ $T = \frac{U - W}{\cos 35^\circ} \quad \text{--- (2)}$ Substitute (2) into (1),	[1] for correct equations by resolution of forces

No.	Solution	Remarks
	$D = (U - W) \tan 35^\circ$ $= [\rho_{\text{sea}} Vg - mg] \tan 35^\circ$ $= [(1.03 \times 10^3)(7.5 \times 10^{-2})(9.81) - (8.0)(9.81)] \tan 35^\circ$ $= 475.7$ $= 476 \text{ N}$	[1] for correct numerical substitution [1] for correct answer
Q(a)(ii)	Work done per unit mass in bringing a small test mass from infinity	[1] for correct